

Notes on Pattern of Snow Accumulation/Ablation on Ice Shelf and Secular Movement of Dakshin Gangotri Glacier Snout in Central Dronning Maud Land, East Antarctica

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Abstract

Six observations were taken on ice shelf near Dakshin Gangotri station, between February 1993 and February 1994. By the end of polar winter in September '93, a net average accumulation of about 50 cm of snow was recorded. The summer of 1993-94 was very mild and the surface recorded a further rise, bringing the total average accumulation to about 68 cm. The resultant rise in the height of the ice shelf would be about 3 cm. The snout of Dakshin Gangotri glacier was compared to its position in 1983. In the past one decade, the glacier tongue has receded by an average of about 7 m.

Introduction

Antarctica is the driest of all continents, where precipitation in the interior is less than 10 cm per annum. But, in the coastal areas, the amount of snowfall is higher and there is also an accumulation by surface drift. Inversely, when the winds are stronger during the blizzards, snow particles are carried away and there is a resultant ablation. Regular periodic observations are needed to find out the exact nature of these accumulation and ablation patterns.

Material and Methods

In the V Indian Antarctic Expedition (1985-86) an area of 100 m by 100 m on ice shelf was selected for such studies (Singh *et al.*, 1988). The area is about 1.25 km east of Dakshin Gangotri station and is free from any obstruction. Here, nine stakes were fixed in a rectangular pattern, at 50 m intervals (Fig. 1). These stakes are being regularly monitored in successive expeditions.

The polar ice sheet has a number of glacier tongues overriding bedrock in Schirmacher range. One such tongue was identified in the II Expedition

(1982-83) and was termed Dakshin Gangotri glacier (Kaul *et al.*, 1985). It is being regularly monitored in successive expeditions to assess oscillations in time and space, which are closely related to internal and external environmental factors.

Results and Discussion

The stakes near Dakshin Gangotri station could be observed only during convoys from Maitri. In one year, between February of 1993 and 1994, six observations were recorded. One stake, number 7, was lost due to high winds and the data are thus limited to eight stakes only. The last observation of November 26, 1992 is taken as a baseline for reference for this study. All these observations are presented in Table 1.

From Table 1, net accumulation or ablation with respect to the preceding observation have been calculated and these are presented in Table 2.

From the above observations, a moderate overall accumulation is deduced for February 1993; ranging from 1 cm to 19 cm. The net average accumulation for this month is 5.9 cm (Fig. 2).

The onset of polar winter coincided with the maximum overall accumulation. Each and every stake recorded a substantial gain, ranging from a minimum of 34 cm to a maximum of 63 cm. The net average accumulation for July 1993 was 46.9 cm (Fig. 3).

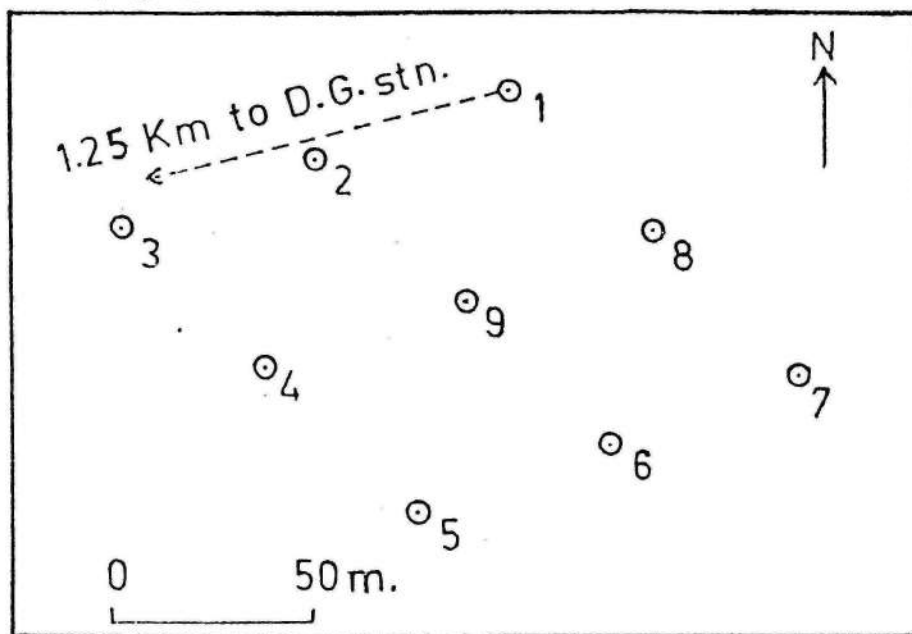


Fig. 1. Location of stakes near Dakshin Gangotri station

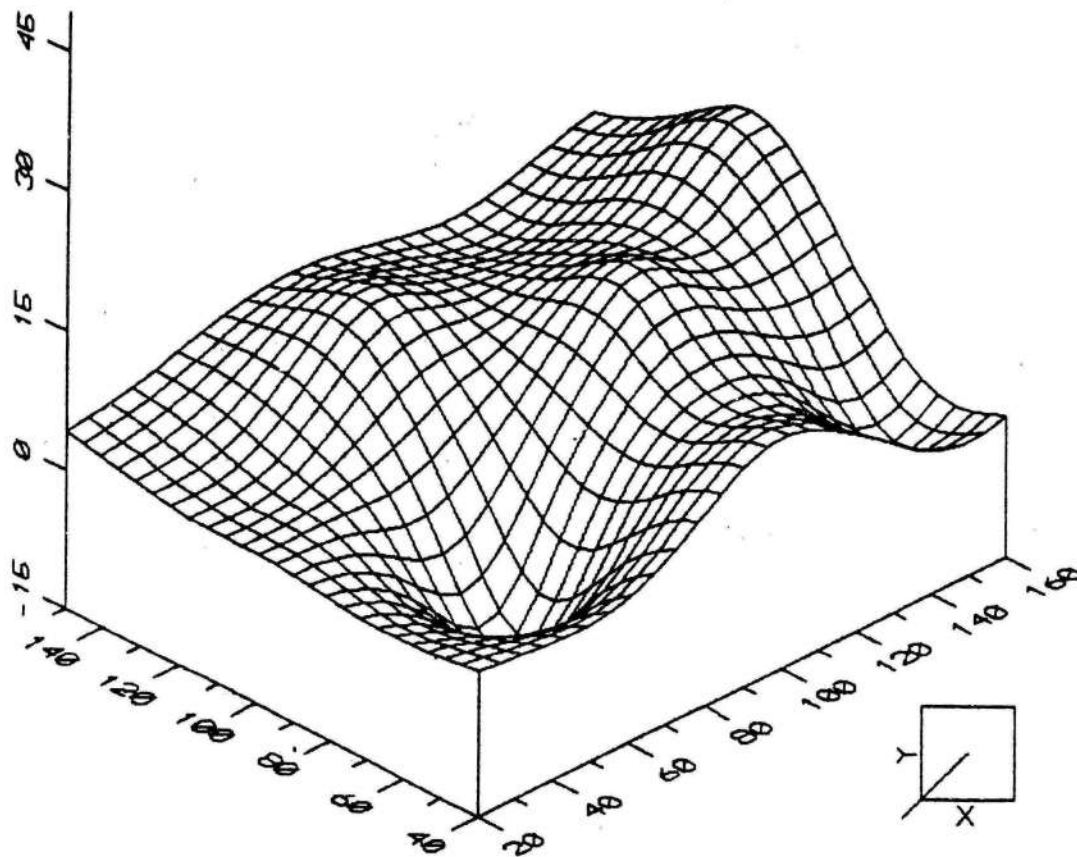


Fig. 2. The accumulation/ablation pattern for the month of February 1993

Table 1 : The height of stakes (in cm) exposed on ice shelf near Dakshin Gangotri station, during 1992-94

Stake No.	Nov.26 1992	Feb.17 1993	Jul.29 1993	Sep.6 1993	Oct.8 1993	Nov.4 1993	Feb.19 1993
1	188	182	127	126	126	114	122
2	188	175	122	126	127	117	110
3	187	186	152	131	132	132	127
4	177	190	146	152	148	144	141
5	182	172	137	123	125	124	118
6	180	184	121	122	126	116	124
8	199	180	139	142	144	116	124
9	197	182	132	128	133	123	116

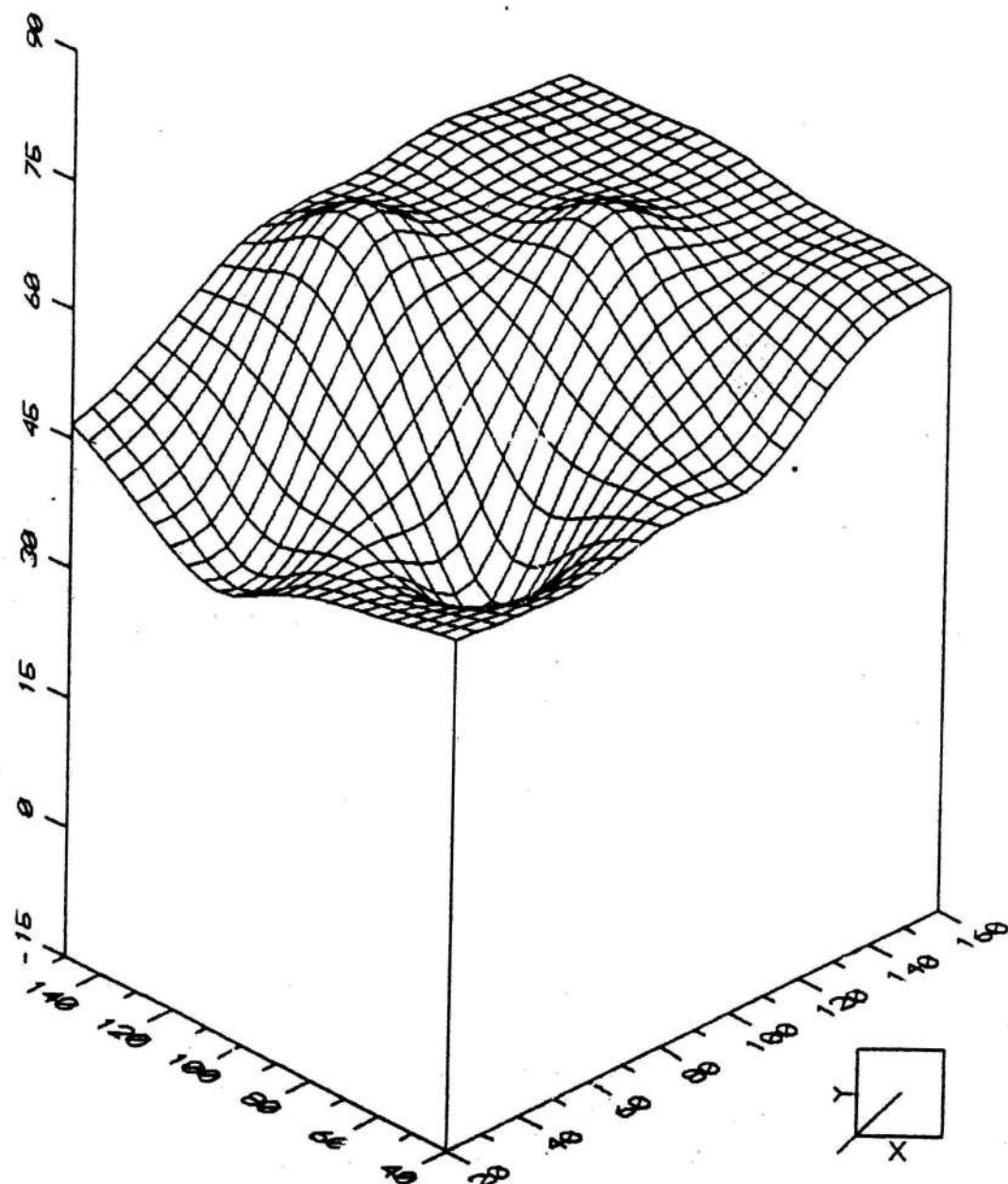


Fig. 3. The accumulation/ablation pattern for the month of July 1993

September was the last month of polar winter. Half of the stakes recorded accumulation upto 21 cm, while the other half showed ablation upto 6 cm. The average net result was an accumulation of 3.3 cm (Fig. 4).

With the beginning of polar summer in October '93, almost all the stakes recorded ablation, ranging from 1 cm to 4 cm. This, incidently, was the only month, which displayed an overall net ablation of 1.4 cm (Fig. 5).

Normally every year, the winter surface of ice shelf peaks around September and then gradually keeps declining through the polar summer months (Ravindra *et al.*, 1994); but the summer of 1993-94 was very mild and the days

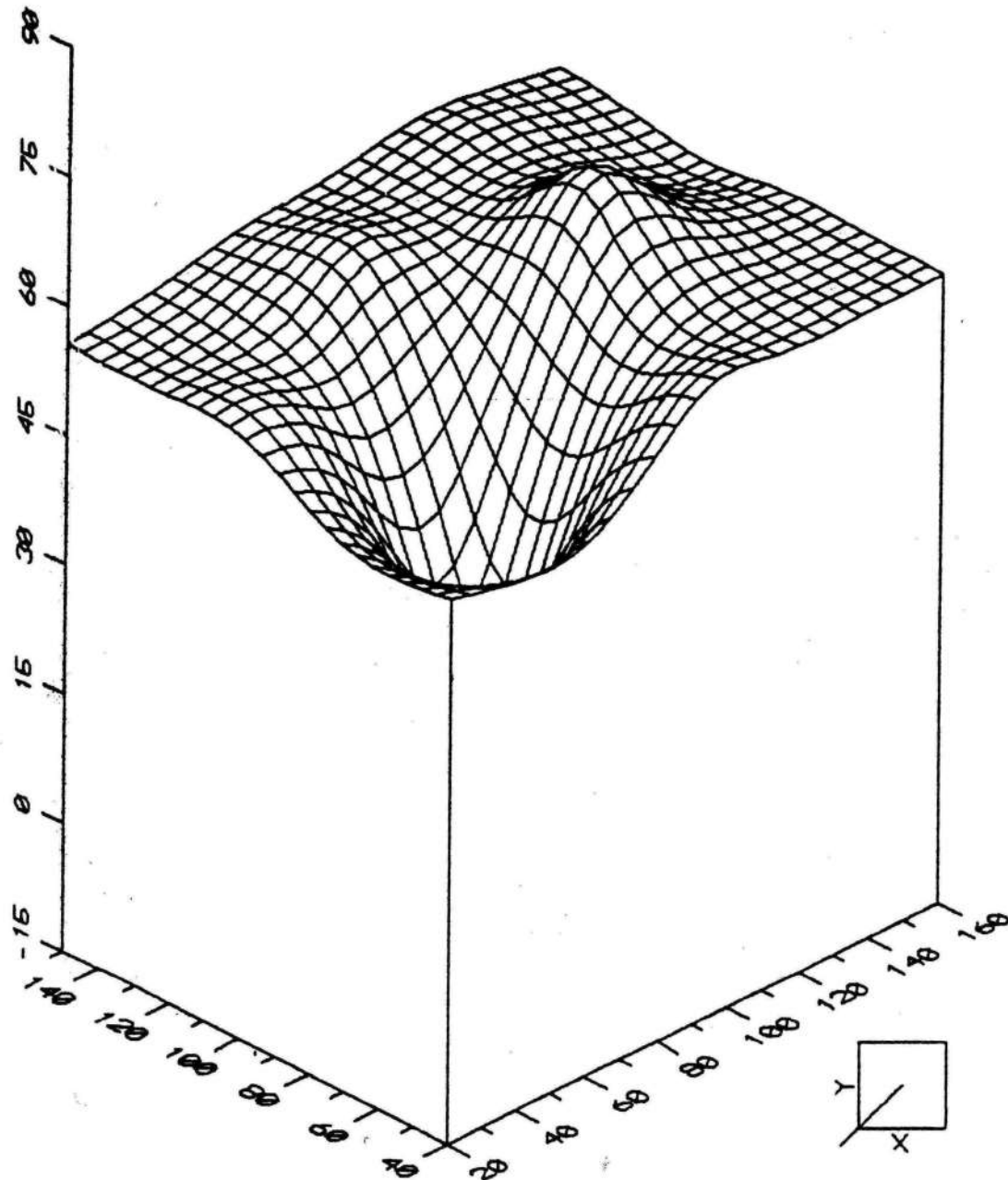


Fig. 4. The accumulation/ablation pattern for the month of September 1993

of high winds were also less. Consequently, the surface kept on rising. The profile of November 1993, shows a net average accumulation of 9.4 cm (Fig. 6); while in February 1994, it has registered a further gain of 4.5 cm (Fig. 7).

By comparing the annual ice shelf surfaces of February 1993 and 1994, a general overall accumulation is observed (Fig. 8). The net average accumulation during this one year is about 68 cm.

The glacier tongue is located in the central part of Schirmacher range ($70^{\circ}45'20''\text{S}$ latitude and $11^{\circ}35'05''\text{E}$ longitude). It is about 6 km west of the Indian station, Maitri. During the XII expedition, the outline of the snout was

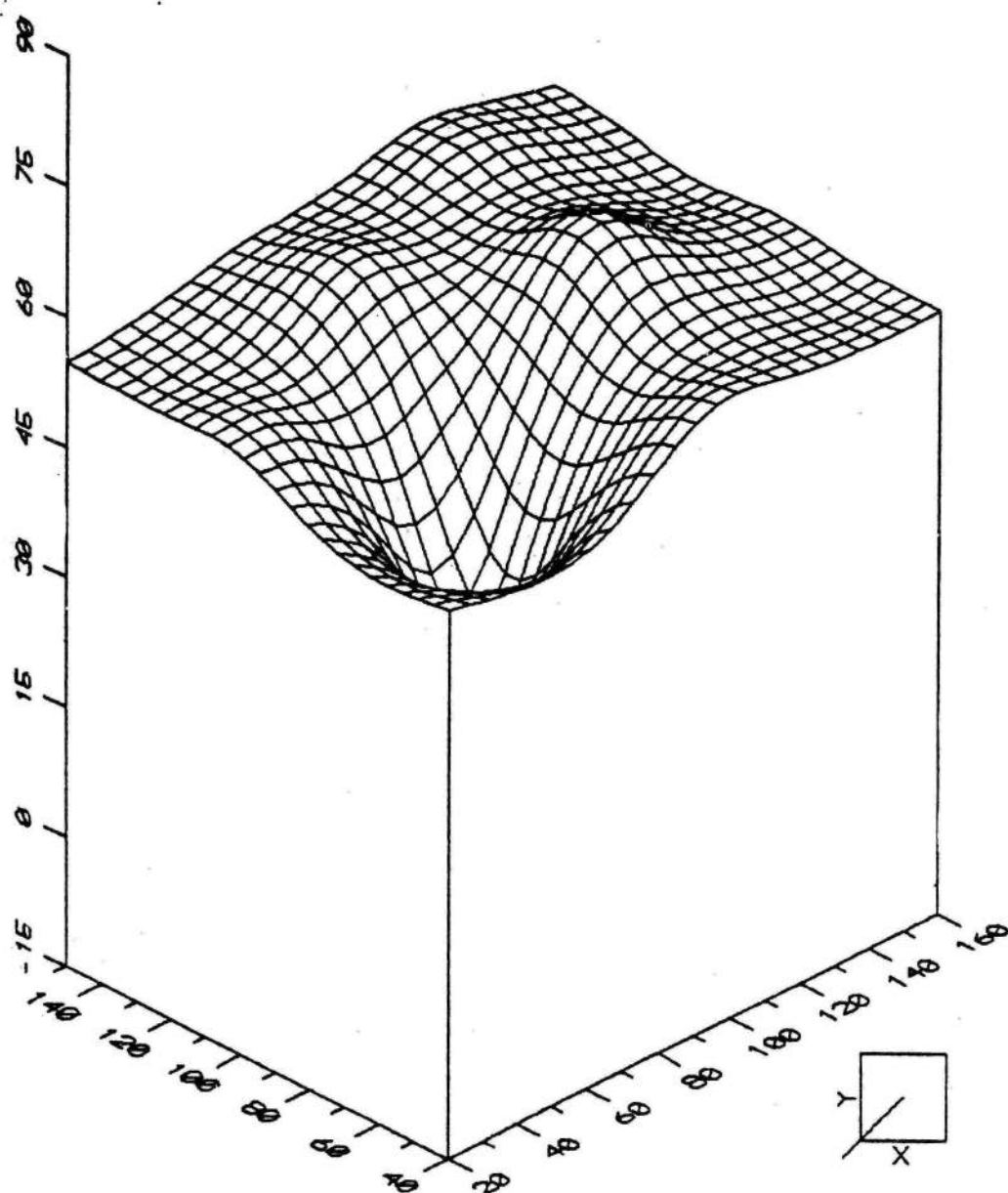


Fig. 5. The accumulation/ablation pattern for the month of October 1993

delineated by tacheometry, in the month of February 1993. Morphologically the tongue has a lobate outline (Fig. 9). The top surface of the glacier has a number of shallow channels, which become active during peak summer months only. This meltwater feeds a small proglacial lake in front of the snout; which overflows to a bigger lake, north of the glacier.

On the basis of the data collected between 1983 and 1987, it was concluded (Kaul *et al.*, 1988) that the northern flank of the glacier has receded in 1986 and 1987. The recession in '87 was in all the sectors of the glacier; but in '86, the right margin of the tongue showed a marked advance. Such oscillatory

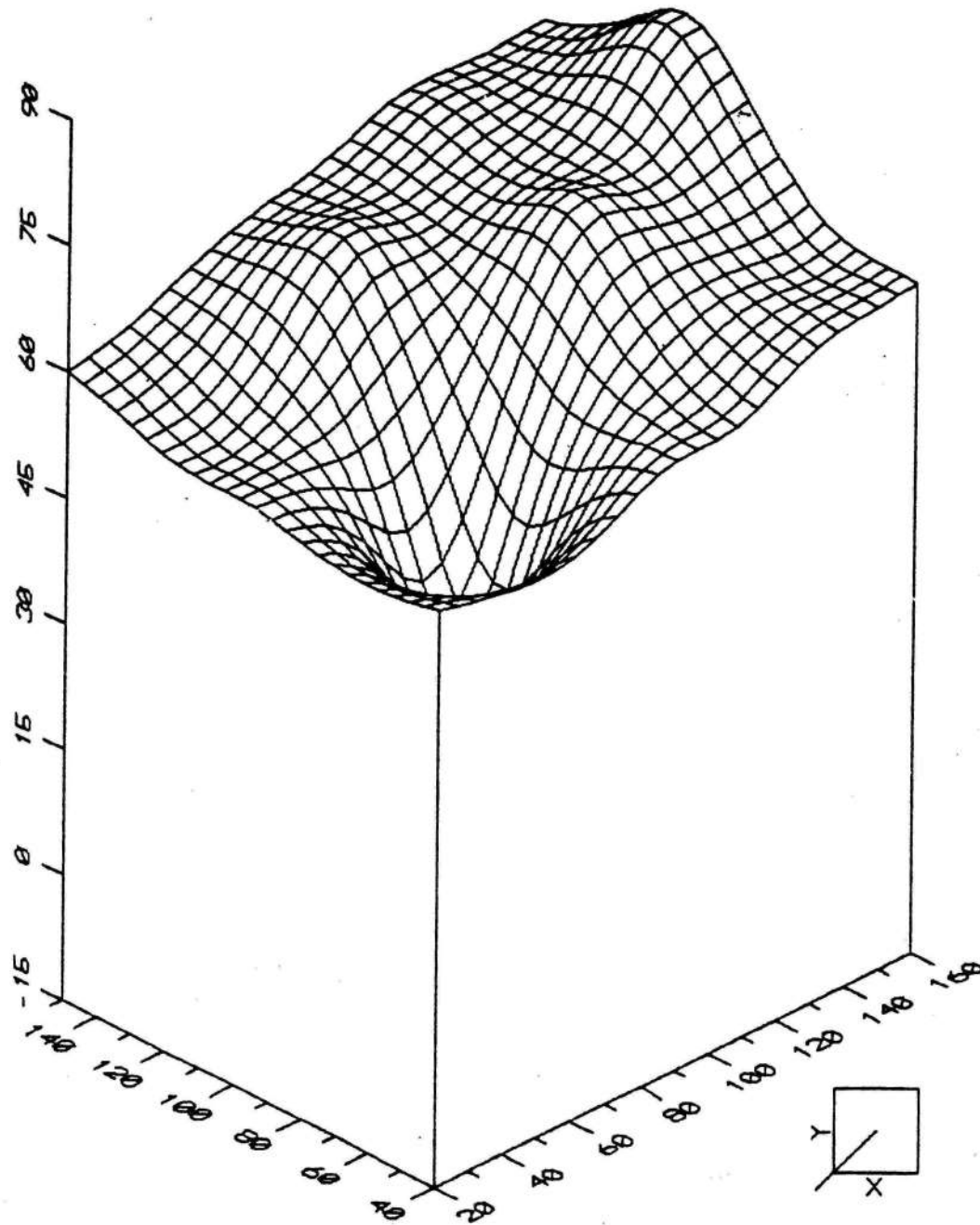


Fig. 6. The accumulation/ablation pattern for the month of November 1993

movement of different parts of the glacier snouts have been recorded in 1991 also (Ravindra *et al.*, 1994).

In the present study, the relative positions of the glacier tongue and proglacial lake in the past one decade have been compared (Fig. 10). It is noticed that from 1983 to 1993, there is a general recession in all the segments of the snout. On an average, the glacial front has receded by 7 m. Correspondingly, the area of the proglacial lake has increased by about 550 sq m.

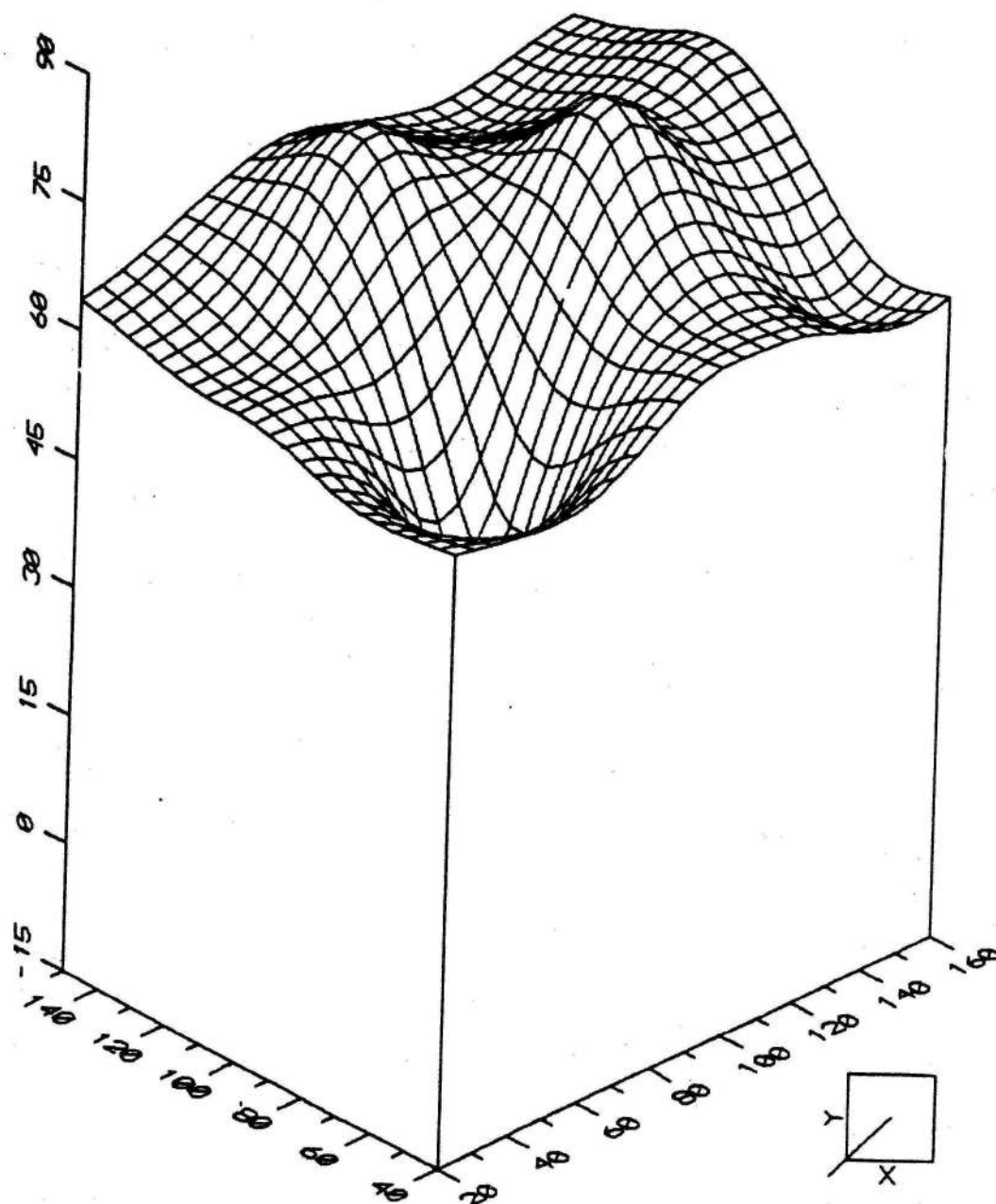


Fig. 7. The accumulation/ablation pattern for the month of February 1994

The ice shelf has recorded a net average accumulation of about 68 cm of snow, between February 1993 and February 1994. After compaction, this would yield about 23 cm of ice. Since, the major part of the floating shelf is submerged below the water-line, it would contribute to about 3 cm net rise in the height of the ice shelf, near Dakshin Gangotri station.

The polar ice sheet, which was overriding Schirmacher range in the past — evident by glacial striations on rocks, polished surfaces and high level morainal deposits — has receded to the present position. Observations since

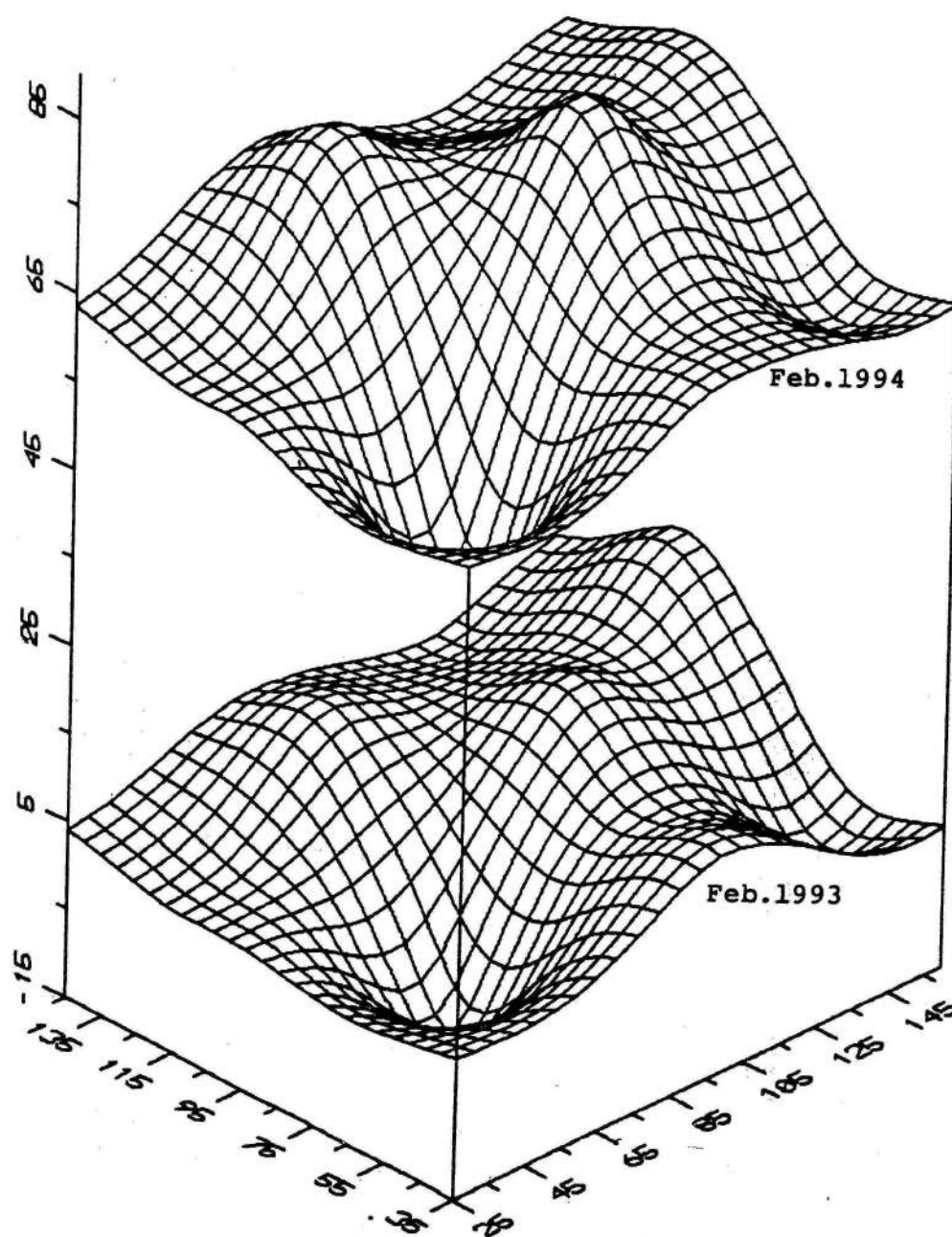


Fig. 8. The accumulation/ablation pattern for the year from February 1993 to February 1994

1983 show that Dakshin Gangotri glacier tongue has recorded an average recession of about 7 m in the past one decade.

Acknowledgements

The authors are grateful to Survey of India team in the expeditions for monitoring the outline of the glacier snout. Thanks are due to the EME team of the X wintering, especially to HMT V. Shinde, for the help in observations on



Fig. 9. The snout of Dakshin Gangotri glacier

Table 2: Net accumulation/ablation of snow (in cm) on ice shelf near Dakshin Gangotri station, during 1993-94

Stake No.	Feb.17 1993	Jul.29 1993	Sep.6 1993	Oct.8 1993	Nov.4 1993	Feb.19 1993
1	+6	+55	+1	0	+12	-8
2	+13	+53	-4	-1	+10	+7
3	+1	+34	+21	-1	+10	+7
4	-13	+44	-6	+4	+4	+3
5	+10	+35	+14	-2	+1	+6
6	-4	+63	-1	-4	+10	+8
8	+19	+41	-3	-2	+28	+8
9	+15	+50	+4	-5	+10	+7

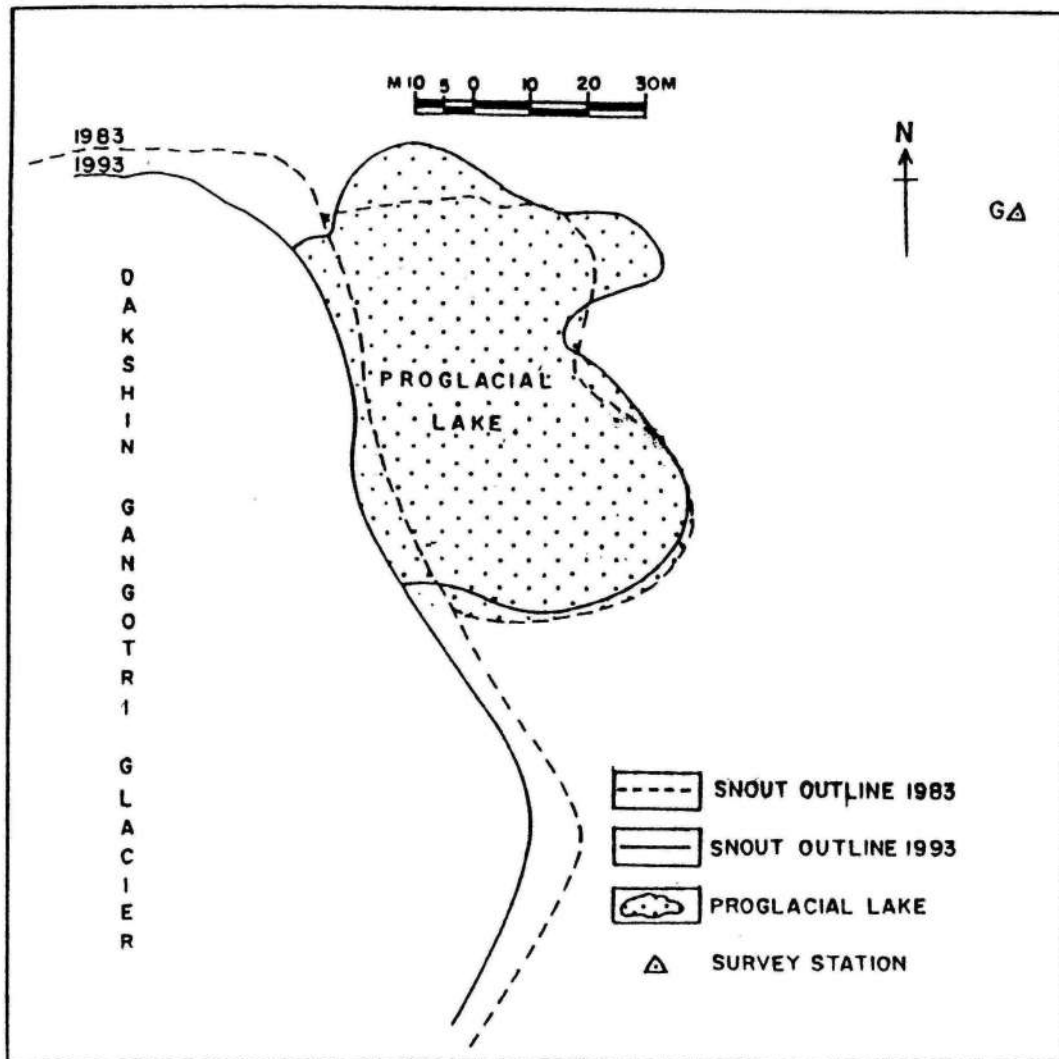


Fig. 10. The position of the glacial front and the area of the proglacial lake in 1983 and 1993

shelf ice. During both the studies, Dr S. Mukerji, Mr S.C. Nautiyal and Mr L.K. Oberoi were associated in February 1993; while Mr M.J. D'Souza & Mr M.J. Beg were field partners in February 1994. The work was carried out under the guidance of Mr M.K. Kaul, Director, Antarctica Division, GSI.

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